

PROJECT 1

SALES DATA SQL ANALYSIS REPORT

Project Overview

This project focused on importing, cleaning, querying, and analyzing a real-world sales dataset using SQL Server Management Studio. The dataset contained **2,824 rows and 25 columns**, and the objective was to develop practical SQL skills in database management, data cleaning, business analysis, and query optimization.

Database Setup

A database named **sales_data** and a table named **sales_data_sample** were created in SQL Server. Appropriate data types such as INT, DECIMAL, VARCHAR, and DATETIME were assigned to ensure data accuracy and integrity.

Data Cleaning and Preparation

Several data quality issues were identified and resolved, including:

- Blank values in STATE, POSTALCODE, and ADDRESSLINE2
- Inconsistent address formatting
- Inconsistent phone formatting
- Improper date formatting in ORDERDATE

SQL functions such as CONVERT() AND ROUND were used to clean and standardize the dataset, improving overall data reliability for analysis.

Business Analysis & Insights

Revenue Performance

- **Total Revenue: \$10,032,628.85**
- **Total Orders: 307**
- **Average Order Value: \$3,553.89**

Sales Trend Analysis

Revenue trends were analyzed across months and years using date-based grouping.

- **2004** recorded the highest annual revenue.
- **November 2004** generated the highest monthly revenue at **\$1,089,048.01**.
- **March 2004** recorded revenue of **\$205,733.73**, showing notable fluctuations and seasonal sales patterns.

Customer Analysis

Customer ranking queries revealed that a small group of customers contributed significantly to company revenue.

Top Customers by Revenue

1. Euro Shopping Channel — **\$912,294.11**
2. Mini Gifts Distributors Ltd. — **\$654,858.06**
3. Australian Collectors, Co. — **\$200,995.41**

Country Performance

The analysis showed that the **USA** generated the highest revenue at **\$3.63M**, followed by **Spain** and **France**, highlighting strong sales concentration in major international markets.

Product Performance

Product line analysis identified the strongest-performing categories:

- **Classic Cars** — **\$3.92M**
- **Vintage Cars** — **\$1.90M**
- **Motorcycles** — **\$1.17M**

Classic Cars emerged as the highest revenue-generating product category, indicating strong customer demand.

Advanced SQL Concepts Applied

The project utilized several advanced SQL techniques, including:

- INNER JOIN used after creating the CUSTOMER TABLE
- GROUP BY and ORDER BY
- Window functions such as RANK() and OVER
- Aggregate functions including SUM() and AVG()

These techniques enabled deeper analytical insights and efficient business reporting.

Query Optimization

Indexes were created on frequently queried columns such as CUSTOMERNAME, PRODUCTLINE AND COUNTRY to improve query performance and retrieval speed.

Example:

```
CREATE INDEX idx_customer  
ON sales_data_sample(CUSTOMERNAME);
```

Conclusion

This project demonstrated how SQL can transform raw business data into meaningful insights through database creation, data cleaning, query development, and analytical reporting. The analysis provided valuable insights into customer behavior, revenue trends, geographic performance, and product demand while strengthening practical SQL Server and data analysis skills.

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